Vitamin A Def

Ophthalmology | 3rd Year MBBS

Hi Sammy!
Vit A deficiency = PSM +
Ophthal cross-topic!
WHO classification = KEY!

Vitamin A | Xerophthalmia | Ophthalmology | 3rd Year

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WHO Classification of Xerophthalmia

WHO classification is
THE most asked table!
Know every stage cold!

DEFINITION

Xerophthalmia = spectrum of ocular manifestations of Vitamin A deficiency, from night blindness to keratomalacia.

WHO Classification of Xerophthalmia (1982)

Stage	Clinical Sign	Key Detail
XN	Night blindness	Earliest symptom, responds to Vit A in 1-2 days
X1A	Conjunctival xerosis	Dry, dull, non-wettable conjunctiva
X1B	Bitot spots	Foamy triangular patches on bulbar conjunctiva
X2	Corneal xerosis	Dry, hazy cornea — reversible with treatment
X3A	Corneal ulcer (<1/3)	Ulceration of <1/3 corneal surface
X3B	Keratomalacia (>=1/3)	Corneal melting — liquefactive necrosis, EMERGENCY
XS	Corneal scarring	End result — permanent vision loss
XF	Xerophthalmic fundus	Peripheral white retinal spots (rare)

★ **X3B (Keratomalacia) = OPHTHALMIC EMERGENCY! Cornea melts within hours-days!**

Stages up to X2 are REVERSIBLE with treatment. X3A/X3B cause permanent damage.

Children don't always progress sequentially — acute illness can cause sudden X3B!

Clinical Features — Stage by Stage

Bitot spots = most asked clinical sign! Know the organism too!

XN — Night Blindness (Nyctalopia)

Earliest manifestation — defective vision in dim light (scotopic vision)

Due to impaired rhodopsin regeneration in rods (Vit A -> retinal -> rhodopsin)

Responds rapidly to Vit A supplementation (within 1-2 days)

X1B — Bitot Spots (Pathognomonic Sign)

Feature	Detail
Appearance	Triangular, foamy, whitish, cheesy patches
Location	Bulbar conjunctiva, temporal side (3 & 9 o'clock)
Composition	Keratinized epithelium + Corynebacterium xerosis (gas-forming)
Foam	Due to gas production by C. xerosis in keratinized debris
Significance	Pathognomonic of Vitamin A deficiency
Response to Rx	Disappears with Vit A therapy (may take weeks)

Corneal Stages (X2, X3A, X3B)

Stage	Finding	Prognosis
X2	Corneal xerosis — dull, hazy, dry cornea	Reversible with urgent Vit A
X3A	Corneal ulcer < 1/3 surface area	Scarring likely, some vision saved
X3B	Keratomalacia >= 1/3 corneal surface	Corneal melting, perforation, blindness

Risk Factors for VAD in Children

- Age: 1-4 years (most vulnerable age group)
- Malnutrition: PEM (kwashiorkor > marasmus for ocular signs)
- Infections: Measles, diarrhea, respiratory infections
- Low dietary intake: poor access to Vit A-rich foods (leafy greens, egg, liver)

WHO dosing schedule = exam GOLD! Must know the exact doses!

Diagnosis of Vitamin A Deficiency

Test	Detail
Serum retinol	< 20 mcg/dL = deficient; < 10 = severe deficiency
Dark adaptation test	Impaired (elevated threshold) — earliest functional test
Conjunctival impression cytology	Loss of goblet cells, keratinization
Relative dose response (RDR)	> 20% = depleted liver stores
Electroretinogram (ERG)	Diminished rod response (scotopic ERG)

WHO Treatment Schedule for Xerophthalmia

Age Group	Dose (Oral)	Schedule
< 6 months	50,000 IU	Day 0, Day 1, Day 14
6-12 months	1,00,000 IU	Day 0, Day 1, Day 14
> 12 months	2,00,000 IU	Day 0, Day 1, Day 14
Pregnant women	10,000 IU/day	For 2 weeks (NOT mega dose!)
Lactating mother	2,00,000 IU	Single dose within 1 month of delivery

★ **NEVER give mega-dose Vitamin A to pregnant women! (Teratogenic risk!)**
Give only 10,000 IU/day for 2 weeks OR 25,000 IU/week for 4 weeks.

Additional Management for Corneal Stages

- Topical antibiotics (moxifloxacin) — prevent secondary infection of ulcer
- Preservative-free lubricants — for xerosis
- Pad and bandage — protect cornea from exposure and desiccation

Prevention + National Programmes

Vit A supplementation programme = PSM cross Q! Know the schedule!

Prevention Strategies (WHO/GOI)

Strategy	Details
Short-term	Vitamin A supplementation (mega-dose) to children + mothers
Medium-term	Food fortification (sugar, oil, flour with Vit A)
Long-term	Dietary diversification — Vit A rich foods in regular diet
Health education	Promote breastfeeding, green leafy vegetables, yellow fruits

National Vitamin A Supplementation Programme (India)

Target:	Children 9 months to 5 years (under-5)
1st dose:	1,00,000 IU at 9 months (with measles vaccine)
Subsequent:	2,00,000 IU every 6 months until 5 years
Route:	Oral (liquid form)
Total doses:	9 doses from 9 months to 5 years
Integrated with:	ICDS (Anganwadi), Pulse Polio, immunization days

Dietary Sources of Vitamin A

Type	Sources
Preformed Vit A (retinol)	Liver, egg yolk, butter, whole milk, fish liver oil
Provitamin A (beta-carotene)	Carrots, papaya, mango, spinach, sweet potato
Richest source	Fish liver oil (cod liver oil)
Richest vegetable source	Carrots, red palm oil, drumstick leaves

★ **WHO Criteria — VAD as Public Health Problem**

Prevalence of Bitot spots > 0.5% OR Night blindness > 1% in children 24-71 months

OR Corneal xerosis/ulcer > 0.01% OR Serum retinol < 20 mcg/dL in > 5% of population

Vit A = cross-topic with
PSM + Paeds + Ophthal!
Triple marks, Sammy!

Most Repeated LAQs (10–12 marks)

1. Describe WHO classification of xerophthalmia. Discuss treatment
2. Write about Vitamin A deficiency — clinical features, prevention, national programme
3. Describe keratomalacia — stages, treatment, and prevention
4. Describe Bitot spots — pathology, clinical features, management

Most Repeated SAQs (5–6 marks)

1. WHO classification of xerophthalmia (table)
2. Vitamin A supplementation schedule for children
3. Bitot spots — pathogenesis and organism involved
4. Dietary sources of Vitamin A
5. Role of Vitamin A in vision (rhodopsin cycle)
6. Prevention strategies for VAD

One-Liner Associations (MCQ/Viva Gold)

Question	Answer
Earliest symptom of VAD	Night blindness (XN)
Pathognomonic sign of VAD	Bitot spots (X1B)
Organism in Bitot spots	Corynebacterium xerosis
Most severe ocular stage	Keratomalacia (X3B)
Vit A dose for child > 1 year	2,00,000 IU on Day 0, 1, 14
Mega-dose contraindicated in	Pregnancy (teratogenic)
Richest dietary source of Vit A	Fish liver oil (cod liver oil)
First dose of Vit A in India	1,00,000 IU at 9 months
Active form of Vitamin A	Retinol (retinal for vision)
VAD most common age group	1-4 years (preschool children)
Vit A stored in	Liver (90% of body stores)

♥ A single Vitamin A dose can prevent a child from going blind, Sammy!
Public health + clinical knowledge = you'll save thousands of eyes!